

10. Electroweak Model and Constraints on New Physics

Revised April 2024 by J. Erler (KPH, JGU Mainz) and A. Freitas (Pittsburgh U.).

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10.1 Introduction

The standard model of the electroweak interactions (SM) [1–4] is based on the gauge group $SU(2) \times U(1)$, with gauge bosons W_μ^i , $i = 1, 2, 3$, and B_μ for the $SU(2)$ and $U(1)$ factors, respectively, and the corresponding gauge coupling constants g and g' . The left-handed fermion fields of the i^{th} fermion family transform as doublets $\Psi_i = \begin{pmatrix} \nu_i \\ \ell_i^- \end{pmatrix}$ and $\begin{pmatrix} u_i \\ d_i \end{pmatrix}$ under $SU(2)$, where $d_i' \equiv \sum_j V_{ij} d_j$, and V is the Cabibbo-Kobayashi-Maskawa mixing [5, 6] matrix. The right-handed fields are $SU(2)$ singlets. From Higgs and electroweak precision data it is known that there are precisely three sequential fermion families. Constraints on V and tests of universality are discussed in Ref. [7] and in Section 12 on the “CKM Quark-Mixing Matrix” in this *Review*. The extension of the formalism to allow an analogous leptonic mixing matrix is discussed in Section 14 on “Neutrino Masses, Mixing, and Oscillations” in this *Review*.

A complex scalar Higgs doublet, ϕ , is added to the model for mass generation through spontaneous symmetry breaking with potential¹ given by,

$$V(\phi) = \mu^2 \phi^\dagger \phi + \frac{\lambda^2}{2} (\phi^\dagger \phi)^2, \quad \phi \equiv \begin{pmatrix} \phi^+ \\ \phi^0 \end{pmatrix}. \quad (10.1)$$

¹There is no generally accepted convention to write the quartic term. Our numerical coefficient simplifies Eq. (10.5a) below and the squared coupling preserves the relation between the number of external legs and the power counting of couplings at a given loop order. This structure also naturally emerges from physics beyond the SM, such as Supersymmetry.

For μ^2 negative, ϕ develops a vacuum expectation value, $v/\sqrt{2} = |\mu|/\lambda$, where $v = 246.22$ GeV, breaking part of the electroweak (EW) gauge symmetry, after which only one neutral Higgs scalar, H , remains in the physical particle spectrum. In non-minimal models there are additional charged and neutral scalar Higgs particles. Higgs boson physics is reviewed in Section 11 on the “Status of Higgs Boson Physics” in this *Review*.

After symmetry breaking the Lagrangian for the fermion fields, ψ_i , is

$$\begin{aligned} \mathcal{L}_F = & \sum_i \bar{\psi}_i \left(i \not{\partial} - m_i - \frac{m_i H}{v} \right) \psi_i - \frac{g}{2\sqrt{2}} \sum_i \bar{\Psi}_i \gamma^\mu (1 - \gamma^5) (T^+ W_\mu^+ + T^- W_\mu^-) \Psi_i \\ & - e \sum_i Q_i \bar{\psi}_i \gamma^\mu \psi_i A_\mu - \frac{g}{2 \cos \theta_W} \sum_i \bar{\psi}_i \gamma^\mu (g_V^i - g_A^i \gamma^5) \psi_i Z_\mu . \end{aligned} \quad (10.2)$$

Here $\theta_W \equiv \tan^{-1}(g'/g)$ is the weak mixing angle and $e = g \sin \theta_W$ is the positron electric charge. Furthermore,

$$A_\mu \equiv B_\mu \cos \theta_W + W_\mu^3 \sin \theta_W , \quad (10.3a)$$

$$W_\mu^\pm \equiv \frac{W_\mu^1 \mp i W_\mu^2}{\sqrt{2}} , \quad (10.3b)$$

$$Z_\mu \equiv -B_\mu \sin \theta_W + W_\mu^3 \cos \theta_W , \quad (10.3c)$$

are the photon field (γ) and the charged ($W^\pm$) and neutral (Z) weak boson fields, respectively.

The Yukawa coupling of H to ψ_i in the first term in $\mathcal{L}_F$, which is flavor diagonal in the minimal model, is $gm_i/2M_W$. From the bosonic interaction Lagrangian,

$$\mathcal{L}_{HV} = \frac{1}{2}(v + H)^2 \left[\frac{g^2}{2} W_\mu^+ W^{\mu-} + \frac{g^2 + g'^2}{4} Z_\mu Z^\mu \right] - V\left(\frac{v + H}{\sqrt{2}}\right) , \quad (10.4)$$

one obtains the EW boson masses (at tree level, *i.e.*, to lowest order in perturbation theory)

$$M_H = \lambda v , \quad (10.5a)$$

$$M_W = \frac{gv}{2} = \frac{ev}{2 \sin \theta_W} , \quad (10.5b)$$

$$M_Z = \sqrt{g^2 + g'^2} \frac{v}{2} = \frac{ev}{2 \sin \theta_W \cos \theta_W} , \quad (10.5c)$$

$$M_\gamma = 0 . \quad (10.5d)$$

The second term in $\mathcal{L}_F$ represents the charged-current weak interaction [8–10], where T^+ and T^- are the weak isospin raising and lowering operators. For example, the coupling of a W to an electron and a neutrino is

$$- \frac{e}{2\sqrt{2} \sin \theta_W} \left[W_\mu^- \bar{e} \gamma^\mu (1 - \gamma^5) \nu + W_\mu^+ \bar{\nu} \gamma^\mu (1 - \gamma^5) e \right] . \quad (10.6)$$

For momenta small compared to M_W , this term gives rise to the effective four-fermion interaction with the Fermi constant given by $G_F/\sqrt{2} = 1/2v^2 = g^2/8M_W^2$. CP violation is incorporated into the EW model by a single observable phase in V_{ij} .

The third term in $\mathcal{L}_F$ describes electromagnetic interactions (QED) [11, 12], and the last is the weak neutral-current interaction [9, 10, 13]. The vector and axial-vector couplings are

$$g_V^i \equiv t_{3L}(i) - 2Q_i \sin^2 \theta_W , \quad (10.7a)$$

$$g_A^i \equiv t_{3L}(i) , \quad (10.7b)$$

where $t_{3L}(i)$ is the weak isospin of fermion i ($+1/2$ for u_i and ν_i ; $-1/2$ for d_i and e_i) and Q_i is the charge of ψ_i in units of e .

The first term in Eq. (10.2) also gives rise to fermion masses, and in the presence of right-handed neutrinos to Dirac neutrino masses. The possibility of Majorana masses is discussed in Section 14 on “Neutrino Mass, Mixing, and Oscillations” in this *Review*.

10.2 Renormalization and radiative corrections

In addition to the Higgs boson mass, M_H , the fermion masses and mixings, and the strong coupling constant, α_s , the SM has three parameters. The set with the smallest experimental errors contains the Z mass², the Fermi constant, and the fine structure constant. The latter two will be discussed in more detail in the next two subsections. (The numerical values quoted in Sections 10.2–10.4 correspond to the main fit result in Table 10.7.)

10.2.1 The Fermi constant

The Fermi constant,

$$G_F = 1.1663788(6) \times 10^{-5} \text{ GeV}^{-2} , \quad (10.8)$$

is derived from the μ lifetime formula³,

$$\frac{\hbar}{\tau_\mu} = \frac{G_F^2 m_\mu^5}{192\pi^3} F(\rho) \left[1 + H_1(\rho) \frac{\hat{\alpha}(m_\mu)}{\pi} + H_2(\rho) \frac{\hat{\alpha}^2(m_\mu)}{\pi^2} + H_3 \frac{\hat{\alpha}^3(m_\mu)}{\pi^3} \right] , \quad (10.9)$$

where $\rho = m_e^2/m_\mu^2$, and where

$$F(\rho) = 1 - 8\rho + 8\rho^3 - \rho^4 - 12\rho^2 \ln \rho = 0.99981295 , \quad (10.10a)$$

$$H_1(\rho) = \frac{25}{8} - \frac{\pi^2}{2} - \left(9 + 4\pi^2 + 12 \ln \rho \right) \rho + 16\pi^2 \rho^{3/2} + \mathcal{O}(\rho^2) = -1.80793 , \quad (10.10b)$$

$$H_2(\rho) = \frac{156815}{5184} - \frac{518}{81}\pi^2 - \frac{895}{36}\zeta(3) + \frac{67}{720}\pi^4 + \frac{53}{6}\pi^2 \ln 2 \\ - (0.042 \pm 0.002)_{\text{had}} - \frac{5}{4}\pi^2 \sqrt{\rho} + \mathcal{O}(\rho) = 6.64 , \quad (10.10c)$$

$$\hat{\alpha}(m_\mu)^{-1} = \alpha^{-1} + \frac{1}{3\pi} \ln \rho + \mathcal{O}(\alpha) = 135.901 . \quad (10.10d)$$

H_1 and H_2 capture the QED corrections within the Fermi model. The results for $\rho = 0$ have been obtained in Refs. [16] and [17, 18] for H_1 and H_2 , respectively, where the term in parentheses is from the hadronic vacuum polarization [17]. The mass corrections to H_1 have been known for some time [19], while those to H_2 are more recent [20]. Notice the term linear in m_e in H_2 whose

²We emphasize that in the fits described in Sec. 10.6 and Sec. 10.7 the values of the SM parameters are affected by all observables that depend on them. This is of no practical consequence for α and G_F , however, since they are very precisely known. Also note that other choices for the three SM input parameters (see *e.g.* Ref. [14]) will lead to equivalent fit results within theoretical uncertainties.

³In the spirit of the Fermi theory, we incorporated the small propagator correction, $3/5 m_\mu^2/M_W^2$, into Δr (see below). This is also the convention adopted by the MuLan collaboration [15]. While this breaks with historical consistency, the numerical difference was negligible in the past.